LLM Processes & the Food Price Report

Forecasting with a language model, and proving it's real

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What we'll cover

01

What is an LLM Process?

Condition on numbers and text; elicit a distribution.

02

Does it work?

Real Food Price Report results vs the classical baselines.

03

The honesty problem

Training cutoffs, leakage, and scoring it fairly.

Can a language model forecast a number?

Not write about forecasting — emit a calibrated distribution over next year's CPI.

> Yes. The interesting question is whether you can believe it.

What is an LLM Process?

Requeima et al., NeurIPS 2024 · 'Context is Key', Williams et al. 2024

A frozen, off-the-shelf LLM — no training, no fine-tuning.

Pack the numeric history into the prompt as text.

Add context: units, domain notes, even a full report

Elicit: a predictive distribution, not a point

Score: head-to-head with ARIMA, same task

It's just a prompt

```
SYSTEM  Probabilistic forecaster.
        Return q05..q95 per step,
        JSON only, monotone.
USER    Task: Food CPI, Jan-Dec 2024
        History:
          2023-05: 185.0
          2023-06: 185.4   # last 30
        Forecast 12 steps, q05..q95.
```

aieng/forecasting/methods/llm_processes/quantile_grid.py

The series is text.

No training, no fine-tuning. We serialize the history, ask for calibrated quantiles, and parse JSON back into a distribution.

Two ways to get a distribution

SAMPLED TRAJECTORIES

Draw N full paths

Quantiles from the samples

Token-heavy: N calls

QUANTILE GRID

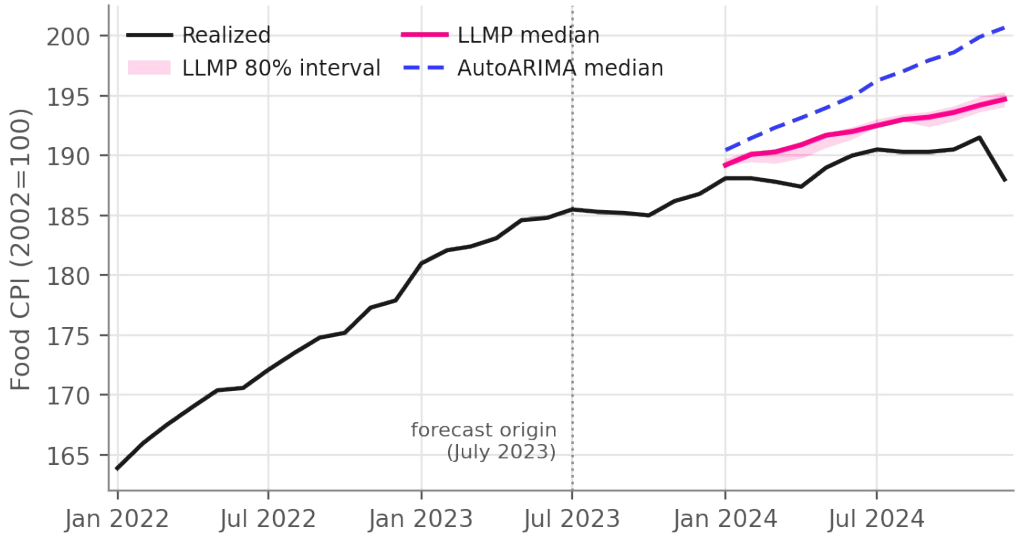
One structured call

Elicit q05..q95 directly

Cheap: context paid once

Same Predictor interface; different elicitation.

What an LLMP forecast looks like



Overall Food CPI · forecast from a July 2023 origin · 80% interval

It tracks the turn.

From July 2023 the LLM-Process forecasts the 2024 plateau and stays close. AutoARIMA just extrapolates the trend upward. Per-origin CRPS 2.5.

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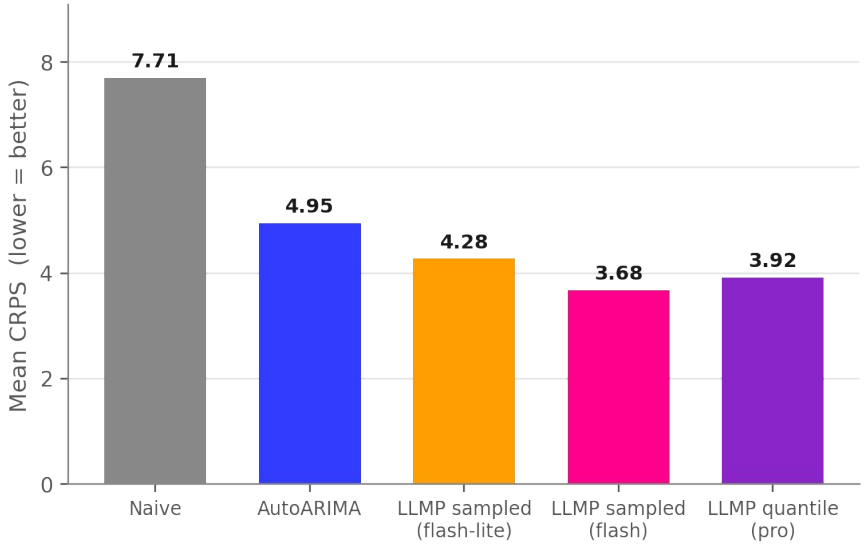
CRPS, a quick reminder

Introduced this morning in Foundations

CRPS scores a whole predictive distribution: the MAE between the forecast CDF and the truth.

It reduces to MAE for a point forecast, and it's strictly proper. Lower is better.

It beats the classical baselines



Mean CRPS across 9 food sub-indices · 6 July origins, 2019-2024

LLMP wins, ~26%.

Mean CRPS: LLMP flash 3.68 vs AutoARIMA 4.95 vs Naive 7.71. Lower is better. A frozen LLM beats a fitted statistical model here.

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THE CATCH

The honesty problem

Why that score is an upper bound

Backtesting an LLM isn't honest

NUMERICAL METHODS

See only data <= origin

Cutoff-safe by construction

Backtest any period

LLMS

Trained to ~early 2025

2019-2024 is in training

Recall, not forecast

Some models even report two cutoffs — stated vs effective.

A historical LLM score is an upper bound.

Best case if the model memorized nothing, and we can't assume that.

> The honest test is a protected, post-cutoff window.

REFERENCE IMPLEMENTATION

The Food Price Report

An annual forecast, reproduced in code

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Canada's Food Price Report

Dalhousie University + partners · dal.ca · annual since 2010



An annual forecast

Next year's food-price inflation, by category — the headline 'groceries up X%'.

Human expertise

Domain analysts and subjective judgment, not a single model.

Evolving methods

Statistics, machine learning, and now LLMs — the techniques change each year.

We reproduce it as a clean, scorable forecasting task.

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The forecast task

Headline avg/avg YoY · LLM-Process vs actual, 2024

Category	LLMP	Actual
Food overall	4.2%	2.7%
Bakery & cereal	4.1%	0.9%
Dairy & eggs	4.3%	2.2%
Restaurants	5.1%	3.6%
Meat	2.8%	3.0%
Vegetables	4.9%	3.6%

Wins on CRPS, yet over-shot the headline.

And still an upper bound — pre-cutoff.

Reports as cutoff-aware context

```
# CFPR PDF -> text + publication_date
doc = extract_document(pdf)
store.add(doc)

llmp = build_llmp_quantile_grid(
    report_sources=["cfpr"],
)
# only reports with
# publication_date <= as_of
```

reports.py · documents/extract.py

Context is Key, literally.

The report is free-text context for the same series. The store only ever shows a report once it was actually published.

What to take forward

01

LLM Processes forecast

A frozen LLM emits calibrated distributions — and wins on CFPR.

02

The score is a ceiling

Pre-cutoff backtests measure recall; trust post-cutoff windows.

03

Context is the frontier

Reports enter the prompt, cutoff-aware. Measuring the lift is open work.

Open 02_food_cpi_experiment.ipynb to run all of this.